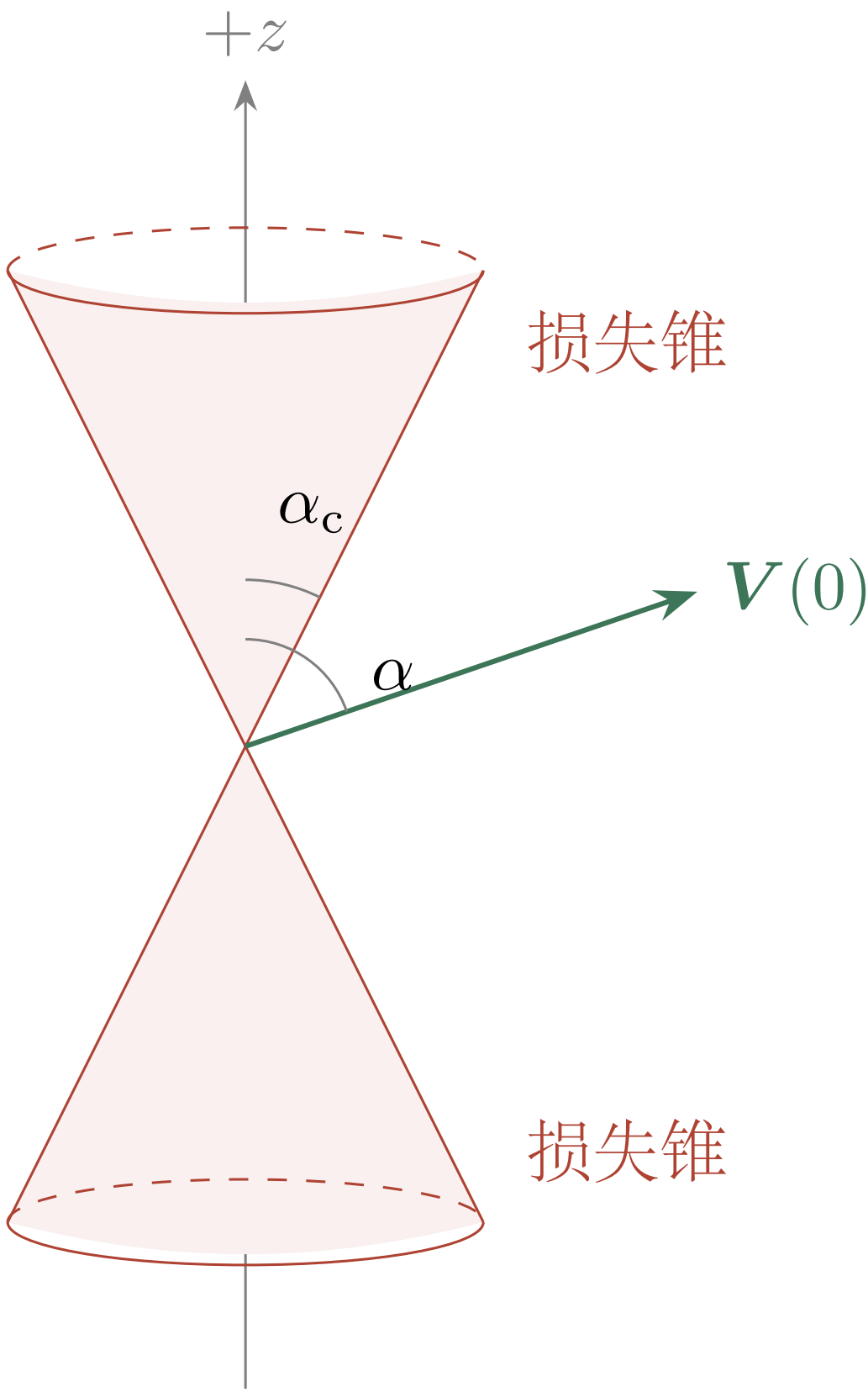

$$\sin^2 \alpha_c = 1/M$$

虚线为锥底背侧；临界等号另作时间检验